Comparative Analysis of the Use of LES and RANS Approaches in 3D Numerical Simulation of Methane Combustion

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***Abstract* — The paper presents the results of numerical simulation of methane combustion for LES and RANS approaches in CFD. CFD modeling was performed using the ANSYS Fluent software. Reynolds-averaged Navier-Stokes equations have been solved using the standard k-ε turbulence model. For the three-dimensional model of large vortices (LES, large eddy simulation), HPC calculations (high performance computing) are used. Simulation of the kinetics of methane-air premixed combustion was performed using the eddy dissipation model (EDC model) for the RANS approach and using the PDF model for the LES approach. To account for radiant heat transfer, the R-1 radiation model was used. An assessment of the possibility of using the vortex dissipation model according to the Damköhler number was made. The scientific substantiation and verification of the physical and mathematical approaches in ANSYS Fluent for the simulation of methane combustion was carried out. A numerical experiment was performed for a turbulent flow. It has been established that in the case of a turbulent flow, the nature of the temperature and velocity contours obtained using the LES and RANS approaches differ significantly. On the basis of a numerical experiment, it was found that the difference between the average temperatures of the combustion products is minimal in the area adjacent to the methane output. The maximum discrepancy of the average temperature is observed for the area of developed turbulence.**

Keywords* — *combustion, methane, numerical simulation, turbulence model, temperature contour

I. Introduction

Numerical simulation of various physicochemical processes in engineering and technology using the specialized software products in recent years has been increasingly used by engineers and scientists from different countries. According to the NASA "Vision 2030" report, by 2030 about 80% of the computational power of all computers will be used for solving problems of numerical simulation of various processes and objects [1].

By now a wide range of solved problems confirms that using special software products such as ANSYS, Comsol, OpenFOAM, etc., makes it possible to solve most of the known scientific and technical problems related to the numerical simulation of various physico-chemical processes. A lot of scientific papers focus on the numerical simulation of the hydrocarbon fuels combustion including the simulation of methane combustion [2], hydrogen [3], synthesis gas [4], etc. The ANSYS software product contains a powerful module with algorithms for calculating the combustion of various fuels, in particular, the combustion kinetics to use models of large eddies (Eddy dissipation model), PDF functions (probability density function) for calculating diffusion burning, and others [5].

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The aim of the study is to compare the results of numerical simulation of combustion of premixed with air methane using the RANS and LES models of turbulence. In addition, a numerical study of the main characteristics of methane combustion based on the developed algorithms was carried out.

The fact of complete or partial replacement of a physical experiment by the numerical simulation depends largely on the accuracy and adequacy of the used models for combustion, turbulence, radiation, etc.

Therefore, one of the objectives of this work is the verification and scientific justification of the ANSYS Fluent models as applied to the CFD simulation of methane combustion.

II. Algorithm Description

Fig.1presents the general scheme for solving the problem of simulating the methane combustion in the form of a flow-chart algorithm.

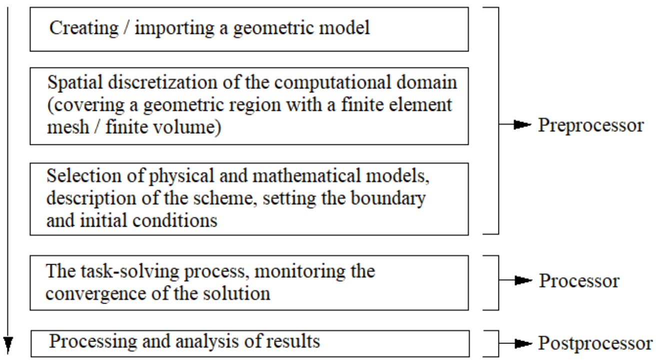

Fig. 1. Flow-chart for solving the problem of methane combustion CFD simulation

To generate the computational domain (calculated geometry), the software Solid Works is used. Generation and grid adaptation is performed in the ANSYS Meshing module. The computational domain and the structure of the computational grid are shown in Fig. 2.

To simulate the flow dynamics in the combustion area, a system of partial differential equations describing the motion of a viscous Newtonian fluid (Navier-Stokes equations) is solved by ANSYS Fluent. Standard form of the Navier-Stokes equations consists of the continuity equation (mass conservation) and the motion equation (momentum conservation) [6].

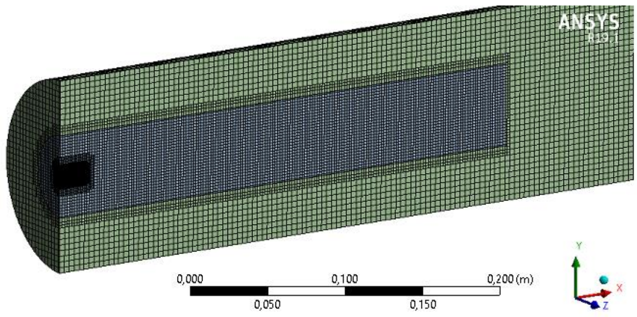

Fig. 2. Computational domain and computational grid structure

The equation of continuity in a standard form for a transient flow can be written as

$$\frac{\partial \rho}{\partial t} + \nabla(\rho \cdot \vec{u}) = 0\,, \quad (1)$$

where u - speed; ρ - density.

The equation of momentum conservation is as follows

$$\frac{\partial}{\partial t}(\rho\vec{u}) + \nabla(\rho\vec{u}\vec{u}) = -\nabla p + \nabla\left(\bar{\bar{\tau}} + \tau'\right) + \rho\vec{g} + \vec{F}\,, \quad (2)$$

where p - static pressure; $\rho\vec{g}$ and $\vec{F}$ - gravitational and external forces, respectively; τ' – Reynolds stress tensor.

The stress tensor in equation (2) can be defined as

$$\bar{\bar{\tau}} = \mu\left[\left(\nabla\vec{u} + (\nabla\vec{u})^T\right) - \frac{2}{3}\nabla\vec{u}I\right], \quad (4)$$

where μ - molecular viscosity; I –unit tensor.

The first term of relation (4) is the strain velocity tensor, and the second term takes into account the volume expansion.

Based on a preliminary analysis and literature review, we used the Redlich-Kwong equation of state in the computational model [7].

Two methods for solving the Navier – Stokes equations were used to simulate combustion: the Reynolds averaged method of the Navier – Stokes equations (RANS) [8] and the large-eddy method (LES) [9, 10]. For the RANS method, the Navier – Stokes equations are solved with the standard k-ε turbulence model (k-ε standard) [11, 12]. The kinetic energy (k) and the dissipation rate of turbulent energy (ε) can be determined from the following relations for the stationary flow:

$$\nabla(\rho \cdot k \cdot \vec{u}) = \nabla\left[\left(\mu + \frac{\mu_t}{Pr_k}\right)\nabla k\right] + G_k + G_b -$$

$$-\rho \cdot \varepsilon - Y_M + S_k\,, \quad (5)$$

$$\nabla(\rho \cdot \varepsilon \cdot \vec{u}) = \nabla\left[\left(\mu + \frac{\mu_t}{Pr_\varepsilon}\right)\nabla \varepsilon\right] + C_{1\varepsilon}\frac{\varepsilon}{k}\left(G_k + C_{3\varepsilon}G_b\right) -$$

$$-C_{2\varepsilon} \cdot \rho\frac{\varepsilon^2}{k} + S_\varepsilon\,, \quad (6)$$

where μ_t - turbulent viscosity; $Pr_{k,\varepsilon}$ - turbulent Prandtl number for k and ε, respectively; $C_{1\varepsilon}$, $C_{2\varepsilon}$, $C_{3\varepsilon}$ - k-ε turbulence model constants; G_k - generation term; G_b - kinetic energy of the buoyant force; Y_M - contribution of variable expansion during compression turbulence to the total dissipation rate (usually taken into account for large Mach numbers).

Constants used in equations (5) and (6) have the following values: $C_{1\varepsilon}$=1,44; $C_{2\varepsilon}$=1,92; $C_{3\varepsilon}$=1; Pr_k=1,0; Pr_ε=1,3.

The LES method is based on the fact that large-scale turbulence is calculated explicitly, and the effects of smaller vortices are modeled using subgrid-closure rules. The conservation equations for modeling large eddies are obtained by filtering instantaneous conservation equations. The flow LES determines the instantaneous position of a "large scale" that resolves the flame front, but the subgrid model requires taking into account the effect of small turbulence scales on combustion. For a jet flame, the LES captures low-frequency parameter changes, unlike RANS, when the result is constant average values. These processes require large computational power. Boundary and initial conditions for the computational domain shown in Fig. 2 are defined as

- outlet section of the air-fuel mixture from the feed channel (inlet section into the combustion area)

$$x = 0;\; T_g = T_{g(in)};\; g_i = g_{i(in)} \quad (7)$$

- outlet section from the combustion area:

$$x = L;\; \frac{\partial g_i}{\partial x} = 0 \quad (8)$$

- center line of the combustion area:

$$y = 0;\; \frac{\partial g_i}{\partial y} = 0;\; \frac{\partial T_g}{\partial y} = 0 \quad (9)$$

Due to the fact that the temperature in the flame front is more than 2000K, and there are the gas stream contains three-atom components, such as H_2O и CO_2, as well as methane, in the combustion area there will be an effect of radiant heat transfer on the enthalpy loss in the gas stream [13]. Despite the fact that some scientists suggest that the effect of radiant heat transfer during methane combustion does not significantly affect the enthalpy loss in the gas flow [14], the model presented by the author uses the radiation model P-1(a simplified version of the more general model P-N) based on the decomposition of the radiation intensity into an orthogonal series of spherical harmonics [15].

When simulating the combustion of methane, for the RANS model we used the global chemical reaction $CH_4+O_2=CO_2+2H_2O$ proposed in the module of the Species Transport of the Fluent solver. The Eddy-dissipation model applied in the calculations is based on the following assumptions: the chemical reaction proceeds fairly quickly with respect to diffusion processes in the reacting flow; when the reactants are mixed at the molecular level, they instantly form reaction products; the reaction rate can be directly related to the time required for mixing the reactants at the molecular level. It is known that such assumptions are valid for the case when the chemical reaction rate is many times greater than the rate of other physical processes, in particular, diffusion. An estimate of the Damköhler number for the methane combustion reaction showed that Da>>10. Therefore, the use of the EDCM model is justified and reasonable [4]. In the case of using the vortex dissipation model, the staged nature of combustion reactions is not taken into account, and the rate of global reactions given above is determined by the time scale of turbulent mixing of the reactants.

PDF model (probability density function). For the PDF model, the two-stage nature of the methane combustion reaction is taken into account [16].

Fluency: second order approximation scheme (second order upwind) [17], number of steps 2000, time of one step 0.001 s, convergence in energy equation 10^{-5}, in equation of momentsпо 10^{-3}.

III. Results and its discussion

The main advantage of CFD-modeling results is their visibility and the possibility of their quick processing and analysis. Fig. 3 shows the simulation results of methane combustion for LES and RANS approaches with an initial methane velocity of 65 m / s and a temperature of 300 K. The temperature of methane and air are equal.

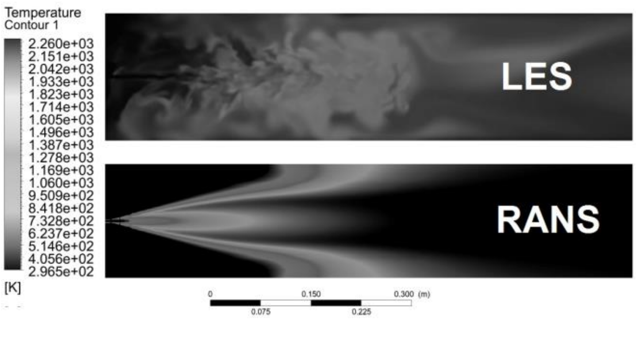

Fig. 3. Results of CFD simulation of combustion using the LES and RANS approaches: temperature contours

Fig. 3 shows significant differences in the nature of the temperature contours obtained using the LES and RANS approaches. So, if for the LES contour of the model, eddies characteristic of turbulent flow are observed, then in the results of the RANS model, the temperature change is smooth. This is due to the fact that the LES approach is based on the fact that large-scale turbulence is calculated explicitly, and the effect of smaller eddies is modeled using subgrid-closure rules. The flow LES determines the instantaneous position of a "large scale" that resolves the flame front, but the subgrid model requires taking into account the effect of small turbulence scales on combustion. For a jet flame, LES captures low-frequency parameter changes, unlike RANS, when the result is constant average values [18]. These processes require large computational power

For clarity, the discrepancies in the results of modeling turbulence (Fig. 4) are shown in the form of velocity contours for the initial methane velocity of 65m / s.

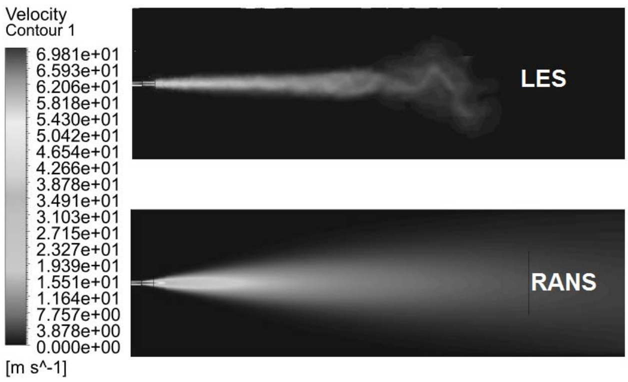

Fig. 4. Speed contours in turbulent flow regime using LES and RANS approach

Results of the LES approach in Fig. 4 show the turbulent eddies during the movement of the fuel-air mixture, while the results of the RANS approach show averaged turbulent eddies and a smooth change in velocity along the combustion axis. However, despite significant discrepancies in the nature of the speed contours, the angle of opening of the jet of the fuel-air mixture for the LES and RANS approaches is the same.

Fig. 5 presents a comparison of an average temperature along the combustion axis of methane obtained using the LES and RANS approaches. The relative distance indicates the ratio of the axial distance to half the total length of the burning area of the combustion z/L. Here, z/L=0 corresponds to the exit of the air-fuel mixture from the inlet nozzle, z/L=1 – to the middle of the burning area. Fig. 5 shows that the maximum discrepancies between the temperatures are observed for the area of the developed flow when turbulent eddies have a significant effect on the heat balance in the combustion area [19]. It should be noted that the temperature differences in the initial section are minimal. For the RANS approach, higher temperatures are observed, since in this case there is no movement of the current lines caused by turbulent eddies.

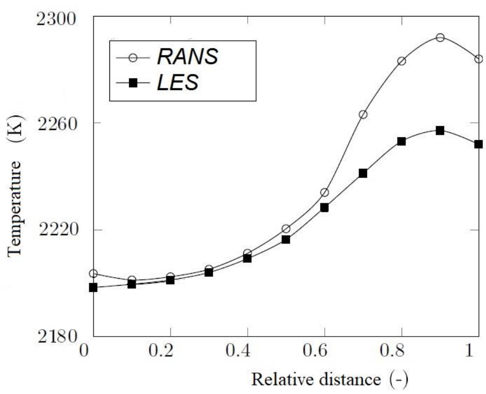

Fig. 5. Comparison of the average temperature over the cross section of the combustion axis for the LES and RANS approaches

IV. Conclusion

This paper presents the results of a numerical study of methane combustion for LES and RANS approaches for modeling fluid flow dynamics. In the LES approach, a PDF-model (probability density function model) of the chemical kinetics of methane combustion is used; in the RANS approach, the eddy dissipation concept model is used. Numerical simulation was performed for the turbulent flow regime with an initial methane velocity of 65 m/s and a temperature of 300K.

The results showed a significant discrepancy in the nature of the contours of temperature and velocity obtained with the LES and RANS approach. Thus, stable turbulent eddies are observed in temperature contours obtained using the LES approach contain, while smooth flow development is observed in the temperature contours obtained using the RANS approach, This is due to the fact that when using the LES approach, large turbulence scales are calculated explicitly, and the effects of smaller vortices are modeled using subgrid closure rules. The conservation equations for modeling large eddies are obtained by filtering instantaneous conservation equations. The flow LES determines the instantaneous position of a “large scale” that resolves the flame front, but the subgrid model requires taking into account the effect of small turbulence scales on combustion. For a jet flame, the LES captures low-frequency parameter changes, unlike RANS, when the result is constant average values. These processes require large computational power.

Comparison of average temperatures in the radial cross sections of the combustion area showed that the maximum discrepancy in the results occurs in the area of developed turbulence. But the temperature discrepancy is minimal in the initial area adjacent to the methane output.

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